

Question # 1: Fill in the blanks:

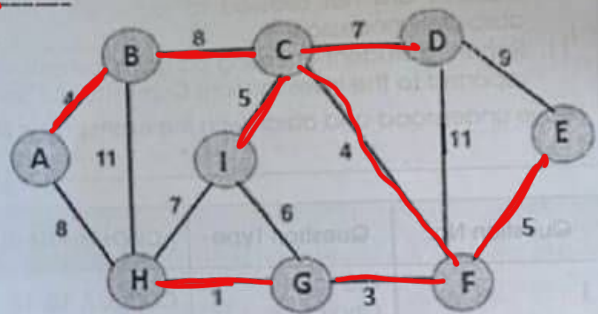
[17]

- If the size of an array is 25, then the The average-case number of basic operation (comparison) done by Sequential search is $\frac{n+1}{2} = \frac{26}{2} = 13$
- If the size of an array is 512, then the The worst-case number of basic operation (comparison) done by binary search is $\log_2(512) + 1 = 10$
- Given the below sorted array:
arr[] = {1, 3, 5, 7, 9, 11, 13, 15}
To determine if x = 10 exists in the above array using the binary search algorithm, the numbers compared in order with x are: 7, 11, 9
- In Mergesort to merge two sub-arrays U = [5, 7, 8, 9] and V = [4, 10, 11, 12] into one array S, how many times the basic operation (comparison) will be executed? 5

*** Answer the following 2 questions using quick sort to sort the following array
S = {5, 1, 7, 9, 3}. Choose the pivot item to be the first item:

- The number of comparisons done by partition function is 4
- In the first step after calling partition function the array S will be: 3, 1, 5, 7, 9

**Answer the following 2 questions by using the following graph:



- Use the Prim's algorithm to find the cost of the minimum spanning tree: 37
- Use the Kruskal's algorithm to find the cost of the minimum spanning tree: 37

*** Suppose there are 4 jobs and the service time of these jobs are $t_1 = 18$, $t_2 = 8$, $t_3 = 12$ and $t_4 = 10$, answer the following 2 questions:

- The job's schedule that minimizes the total time spent in the system is: 2, 4, 3, 1

- The total time spent in the system for the optimal schedule is 104

- The number of the basic operation (comparison) to find the Longest Common Sub-sequence of the following 2 sequences: "President", "Providence" is: 90

- Based on the information given in the next table, which of the following could be a feasible sequence generated when solving the job scheduling with deadlines problem using the greedy method?

Job	Deadline	Profit
1	1	100
2	2	75
3	1	80
4	3	150
5	2	60

1, 2, 4

**Answer the following 2 questions by using the dynamic programming approach to compute $\binom{5}{3}$

- Compute the value of binomial coefficient $\binom{5}{3}$ 10

- How many minimum terms that will be computed to find the value of $\binom{5}{3}$ 18

9. The big theta of $F(n) = 6F\left(\frac{n}{2}\right) + n^3$

A) $n^2 * \log n$ B) n^2 C) $n^3 * \log n$ D) n^3

10. Dijkstra's algorithm is used for Finding :

- A) The minimum spanning tree. B) The Longest common sub-sequence.
C) The shortest path between every pair of vertices in a graph.

D) The shortest paths from a single vertex to every vertex in a graph.

11. Prim's algorithm is a

A) Divide and conquer algorithm

B) Greedy algorithm

C) Dynamic programming algorithm

D) Backtracking algorithm

12. Which of the following algorithm has the highest time complexity

A) Floyd's

B) Prim's

C) Kruskal's

D) Dijkstra's

13. Which of the following encoding is an example of binary prefix-free code?

A) a: 1, b: 10, c: 00, d: 011

B) a: 00, b: 01, c: 1, d: 111

C) a: 1, b: 01, c: 001, d: 000

D) a: 10, b: 00, c: 01, d: 11

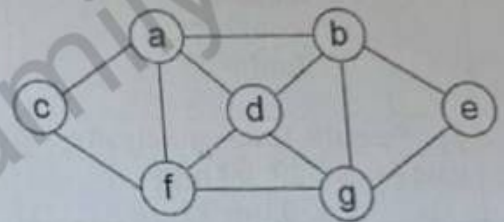
Which of the following is a possible Hamiltonian circuit of the graph on the left?

A) {c, a, b, d, f, g, e}

B) {c, a, b, e, g, f, c}

C) {c, a, d, b, e, g, d, f, c}

D) {c, a, d, b, e, g, f, c}



Depth First Search is equivalent to which of the traversal in the Binary Trees?

A) Level-order

B) Post-order

C) Pre-order

D) In-order

The number of nodes in the state space tree without pruning to the sum-of-subsets problem using backtracking is

A) n^n B) $\frac{n^{n+1} - 1}{n - 1}$ C) $n!$ D) $2^{n+1} - 1$

In n-queens problem, by taking the advantage that no 2 queens can be placed in the same row or in the same column, the number of candidate solutions in the state space tree without pruning is:

A) n^n B) $\frac{n^{n+1} - 1}{n - 1}$ C) $n!$ D) $2^{n+1} - 1$

***** Write your answers of the Question # 1 & #2 in the table below:

Question # 1	Q1	Q2	Q3	Q4	Q5	Q6	Q7	Q8
Q9	Q10	Q11	Q12	Q13	Q14	Q15	Q16	Q17

Question # 2	Q1	Q2	Q3	Q4	Q5	Q6	Q7	Q8
Q9	Q10	Q11	Q12	Q13	Q14	Q15	Q16	Q17



Question # 3: Based on the information of the matrix below, which represents the matrix generated by Floyd's algorithm at the 1st iteration (i.e. $D^{(1)}$) to find shortest paths between every pair of vertices in a specific weighted digraph, find the matrix which will be generated at the 2nd iteration (i.e. $D^{(2)}$). [4 pts]

$$D^{(1)} = \begin{bmatrix} 0 & 3 & 9 & 10 \\ 2 & 0 & 3 & 4 \\ 4 & 1 & 0 & 4 \\ 7 & 3 & 8 & 0 \end{bmatrix}$$

$$D^{(2)} = \begin{bmatrix} 0 & 3 & 6 & 7 \\ 2 & 0 & 3 & 4 \\ 3 & 1 & 0 & 4 \\ 5 & 3 & 6 & 0 \end{bmatrix}$$

Question # 4: Given the two sequences "MONPMN" and "MPNMMN0", complete the table below to find the Longest Common Sequence (LCS) using Dynamic Programming. Then determine the traced locations by backtracking algorithm. [4 pts]

j		0	1	2	3	4	5	6
i			M	O	N	P	M	N
	0	0	0	0	0	0	0	0
1	M	0	1	1	1	1	1	1
2	P	0	1↑	1↑	1↑	2	2	2
3	N	0	1↑	1↑	2	2↑	2↑	3
4	M	0	1↑	1↑	2↑	2↑	3	3↑
5	M	0	1↑	1↑	2↑	2↑	3	3↑
6	N	0	1↑	1↑	2	2↑	3↑	4
7	0	0	1↑	2	2↑	2↑	3↑	4↑

MPMN or MNMN

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After creating pruned state space tree by using backtracking to solve the sum of subset problem with $n = 3$, $W = 10$, and $w_1=3$ $w_2=5$ $w_3=7$ answer the following 2 questions

16. The values of leaves in the pruned free space tree from left to right are: _____

17. The number of promising nodes in the pruned free space tree is _____

18. In m-coloring problem if there is a graph with 5 vertices and 4 colors ($m = 4$), the number of nodes in the second level of the free space tree without pruning is _____

Question # 2: Choose the correct answer:

[17 x 1pt]

1. Find the time complexity of this function .

```
void fun(int n) {
    if (n==1) {
        cout<<"*";
    }
    else {
        cout<<"*";
        cout<<"*";
        fun(n-1);
    }
}
```

A) $T(n) = T(n-1) + 2$ $T(1)=1$;

B) $T(n) = T(n-1) + 1$ $T(1)=2$;

C) $T(n) = 2T(n-1) + 2$ $T(1)=0$;

D) $T(n) = T(n) + 3$ $T(1)=0$;

2. Find the time complexity of print function .

```
void fun(int a[], int m) {
    for (i=0; i<m*2; i++)
        cout<<a[i];
}

int print (int a[], n) {
    fun(int a[], n/2);
    for (i=1; i<n*3; i++)
        cout<a[i];
}
```

A) $3n - 1$

B) $4n - 1$

C) $5n - 1$

D) $6n - 1$

$m = \frac{n}{2}$

$2m + 3n - 1 \rightarrow 4n - 1$

3. Which of the following is the correct order of growth (from the lowest to highest)?

A) $\theta(n^2)$, $\theta(2^n)$, $\theta(n!)$, $\theta(n^n)$

B) $\theta(n^2)$, $\theta(n!)$, $\theta(2^n)$, $\theta(n^n)$

C) $\theta(2^n)$, $\theta(n^2)$, $\theta(n^n)$, $\theta(n!)$

D) $\theta(2^n)$, $\theta(n^2)$, $\theta(n!)$, $\theta(n^n)$

4. The time complexity of Fibonacci algorithm using dynamic programming is:

A) $\theta(\lg n)$

B) $\theta(n)$

C) $\theta(n^2)$

D) $\theta(2^n)$

5. Which sorting algorithm has the least worst case time complexity

A) Exchange sort

B) Merge sort

C) Quick sort

D) None

*** Answer the following 2 questions for the given recurrence equation below:

$t_n - 4t_{n-1} + 3t_{n-2} = 0$ $t_0 = 0$, $t_1 = 2$

6. What is the characteristic equation?

A) $r^2 + 4r - 3 = 0$

B) $r^2 - 4r + 3 = 0$

C) $1 + 4r - 3r^2 = 0$

D) $1 - 4r + 3r^2 = 0$

7. What is the particular form of the solution of the recurrence equation?

A) $t_n = (3)^n - 1$

B) $t_n = (3)^n + 1$

C) $t_n = (-3)^n + 1$

D) $t_n = (-3)^n - 1$

8. The solution of the following recurrence is: $T(n) = T(n-1) + 4n$, $T(1) = 6$

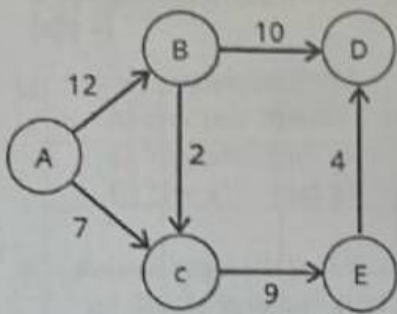
A) $4 * n + 2$

B) $6 * n - 2$

C) $2 * n^2 + 2 * n + 2$

D) $3 * n^2 + 3 * n - 2$

Question # 5: Based on the information given in the below graph, complete the below table after applying Dijkstra's Algorithm starting from vertex (A) [4 pts]

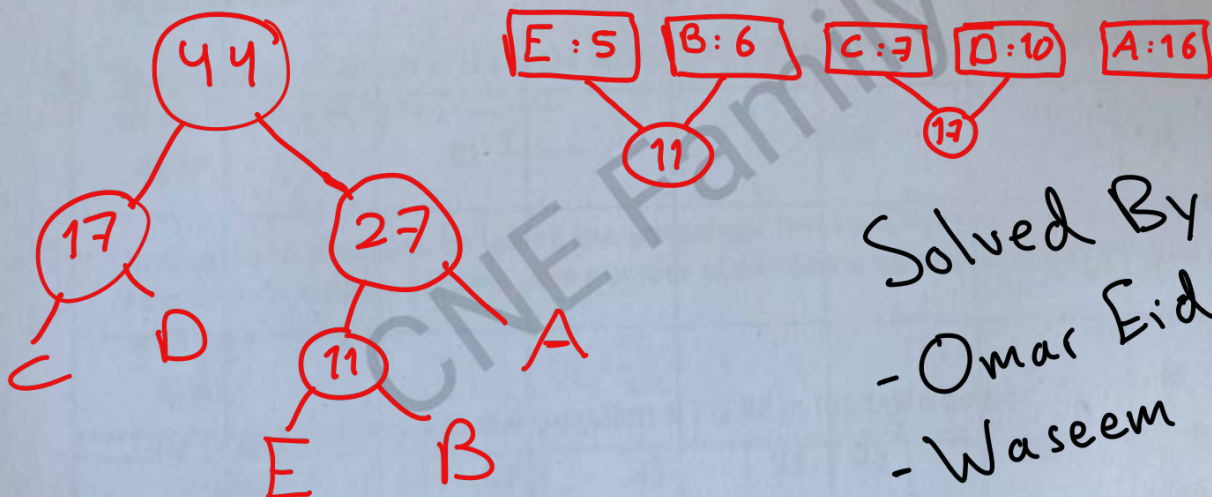


(vnear)	(length)			
Selected vertex	B	C	D	E
A	12	7	∞	∞
C	12	7	∞	16
B	12	7	22	16
E	12	7	22	16
D	12	7	20	16

Question # 6: Answer the following questions supposing a data file has characters and frequencies shown in the table below. If Huffman coding is used: [4 pts]

Character	Frequency
A	16
B	6
C	7
D	10
E	5

A- Construct the Huffman tree of the file (input) characters



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B- Fill in the table below with the code-word for each character.

Character	Code
A	
B	
C	
D	

A=11
B=101
C=00
D=01
E=100